ICE SUGAR NO. 11 FUTURES (SB)

Spread & Curve Structure — Fundamental Reference

Global Production, Trade Policy, Ethanol, Weather, and the Forward Curve

This document presents a fact-based reference on the fundamental drivers of ICE Sugar No. 11 futures (ticker: SB), the global benchmark for raw (unrefined) cane sugar traded on the Intercontinental Exchange (ICE Futures US). Sugar No. 11 is one of the oldest and most globally significant agricultural futures contracts. It prices the international trade market for raw sugar on a free-on-board (FOB) basis at the port of origin, in contrast to Sugar No. 16 (the domestic US market) or the ICE White Sugar No. 5 (London, refined sugar). The market is shaped by a unique combination of factors: a highly politicised global trade environment (government price supports, import quotas, and export subsidies distort production in many countries); the world's dominant producer Brazil, whose sugarcane processors choose between making sugar and ethanol based on relative prices; monsoon-dependent production in India and Thailand; and a demand structure that is growing in emerging markets but subject to health policy headwinds in developed markets. Understanding all of these dimensions across the forward curve is the foundation of spread and butterfly trading in this market.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Settlement Mechanics

Sugar No. 11 is a physically delivered contract. The grade and delivery terms are precisely specified to ensure the contract reflects the international free market for raw cane sugar — explicitly excluding the heavily protected US domestic market.

Parameter	Detail
Exchange	ICE Futures US (formerly NYBOT — New York Board of Trade)
Ticker	SB (front month)
Full name	Sugar No. 11 (World Raw Sugar)
Contract size	112,000 pounds (approximately 50.8 metric tonnes)
Quotation	US cents per pound (¢/lb); minimum tick = 0.01¢/lb = $11.20 per contract
Contract months	March (H), May (K), July (N), October (V) — four delivery months per year. Active contracts extend approximately 2–3 years forward. Liquid trading concentrated in front 3–4 contracts.
Settlement type	Physical delivery — raw cane sugar, FOB stowed at named ports in the country of origin. The seller designates the delivery port from the list of approved origins (approximately 30+ countries eligible). Buyer receives a warehouse receipt or shipping document.
Delivery grade	Raw cane sugar, 96 degrees polarisation minimum (pol — a measure of sucrose purity). Higher pol earns a premium; lower pol (minimum 95.4 pol) earns a discount. Moisture content, grain size, and colour specifications also apply.
Last trading day	Last business day of the month preceding the delivery month
Trading hours	ICE electronic: Sunday–Friday 02:30–14:00 ET; liquid hours approximately 07:00–14:00 ET
Contract value at 20¢/lb	$22,400 per contract
Price limits	No hard price limits; ICE uses dynamic circuit breakers
Relationship to other sugar contracts	Sugar No. 16 (SBV): domestic US raw sugar, higher-priced due to import protection. ICE White Sugar No. 5 (London): refined white sugar ($/MT). The White Premium (No. 5 minus No. 11 in equivalent units) measures refinery margins globally.

1.1 The 96-Degree Polarisation Standard

- "Pol" or degree of polarisation measures the sucrose content of raw sugar using optical rotation of polarised light. 96 pol means approximately 96% sucrose purity — the standard raw sugar delivered against the No. 11 contract. Raw sugar at this specification is the feedstock for refineries that produce white (refined) sugar. The 4% impurities (molasses, colour, moisture) are removed in the refining process.
- The pol specification creates a basis structure: sugar above 96 pol earns a delivery premium (the physical market pays more for higher purity). Sugar between 95.4 and 96 pol is deliverable at a discount. The range of polarisations delivered against the contract affects the convergence of futures to physical cash prices.
- The "White Premium" (white sugar price minus raw sugar price, expressed in $/MT or ¢/lb equivalent) is the market's measure of refinery margin. Wide white premium = strong refinery economics; narrow = refinery margins compressed. The white premium is closely tracked by producers who can choose to refine before export.

1.2 The FOB Delivery Structure

- Delivery is FOB (Free On Board) at the loading port, meaning the seller bears all costs until the sugar is loaded on the vessel. The buyer bears freight costs from the origin port to the destination. This FOB structure means that No. 11 prices do not include freight — the landed cost of sugar in a consuming country is No. 11 plus the freight differential from the origin.
- The wide range of eligible delivery origins (Brazil, Colombia, Cuba, Guatemala, El Salvador, Honduras, Nicaragua, Philippines, South Africa, Swaziland/Eswatini, Thailand, and others) means the cheapest deliverable origin typically determines the futures price level. The market prices the "cheapest to deliver" (CTD) origin, which varies with relative production costs, freight differentials, and currency exchange rates.

2. Global Production — The Supply Landscape

World raw sugar production is approximately 180–200 million metric tonnes per year (raw value). Supply is geographically concentrated in a small number of dominant producers whose output decisions, weather events, and policy choices determine the global balance and the No. 11 price.

Country / Region	Production (approx.)	Crop Type	Crop Calendar	Key Risk	No.11 Curve Impact
Brazil (Centre-South)	35–40 MMT	Sugarcane	CS harvest: Apr–Nov. Off-season: Dec–Mar.	Drought (ENSO); frost risk (Jul–Aug); sugar vs ethanol allocation; BRL/USD rate	Dominant — Brazil drives the global balance. CS harvest pace is the primary monthly supply signal. Ethanol parity is the critical micro-structural variable.
Brazil (North-East)	2–3 MMT	Sugarcane	Oct–Mar harvest (counter-seasonal to CS)	Drought (NE Brazil is semi-arid); logistics (port congestion)	Minor volume but sometimes relevant when CS harvest disappoints. Counter-seasonal timing matters.
India	30–36 MMT	Sugarcane	Oct–Mar crushing. Maharashtra, UP, Karnataka.	Monsoon (ENSO: La Nina = good; El Nino = risk); government intervention (MSP, export quotas); cane area	Second-largest producer. India's export policy (government approval required) is binary and market-moving when announced. Swing factor for the global balance.
EU	15–18 MMT	Sugar beet (Sep-Dec harvest)	Beet harvest Sep–Dec; campaign ends Dec–Jan.	Drought (2018, 2022 reduced EU beet yields sharply); cercospora beet leaf spot disease; neonicotinoid ban effects on beet establishment	Significant but protected market. Internal EU prices governed by beet contracts. EU export volumes (white sugar) affect No.11 via white premium.
Thailand	8–12 MMT	Sugarcane	Nov–Apr crushing season.	Drought (ENSO: El Nino reduces yields); cane area competition from cassava; government cane price policy	Third-largest exporter. Thai crop size directly affects Asian regional prices and No.11 premium/discount. Tends to peak or disappoint in ENSO years.
China	10–12 MMT	Sugarcane (Guangxi) + beet (NE China)	Cane: Nov–Apr. Beet: Sep–Nov.	Drought; pest/disease in Guangxi; government import quota policy; substitute imports (HFCS from US corn)	Large import demand (4–6 MMT/yr) makes China a swing buyer that can tighten No.11 when quotas expand. Import quota announcements are market-moving.
Pakistan	6–8 MMT	Sugarcane	Nov–Feb crushing.	Water availability (Indus irrigation); government minimum support prices; smuggling to/from India	Self-sufficient in most years; occasionally a net exporter or importer depending on balance. Less directly relevant to No.11 but adds to global surplus/deficit.
Australia	4–5 MMT	Sugarcane	Jun–Dec harvest (Queensland).	La Nina: heavy rainfall, wet harvest, lower extraction rate; cyclone risk; El Nino: drought reduces yields	Significant exporter; counter-seasonal to NH; Australian crop provides a southern hemisphere supply signal during May–Oct No.11 contracts.

Mexico, Colombia, Guatemala, Cuba	2–6 MMT each	Sugarcane	Varies by country (Jan–Jun typical for Central America)	Weather; domestic fuel ethanol policy (Mexico); political/logistical factors	Collectively important as export origins. Cuban sugar has declined from historical peaks but remains a deliverable origin.

The global sugar market operates on an October-September crop year (aligned roughly with the Northern Hemisphere harvest). The ISO (International Sugar Organization) and USDA track global production, consumption, and trade on this basis. The market distinguishes between production years (the growing and crushing season) and trade years (the marketing of that production, which may lag production by months). Brazil's Centre-South harvest running April–November does not align with the October-September crop year — a source of analytical complexity.

3. Brazil — The Dominant Supply Variable

Brazil's Centre-South (CS) sugarcane region is the single most important supply variable in the global sugar market. No other commodity market is so thoroughly dominated by a single producer: Brazil represents approximately 20–25% of world production and 40–50% of world sugar exports. Every aspect of the Brazilian CS harvest is closely monitored by the market.

3.1 The Centre-South Harvest Cycle

- **Harvest season: April–November**. Crushing typically begins in April (southern mills) and accelerates through May–October, peaking in August–September. The harvest ends in November as rainfall returns (rain prevents field operations and dilutes cane sucrose content). The off-season (December–March) sees zero crushing in CS.
- **Cane ATR (Total Recoverable Sugar)**: a measure of the sucrose and fermentable sugars available per tonne of cane. ATR peaks in the dry season (July–September) when cane matures, and is lower early and late in the season. Poor ATR (due to excessive early rain, drought stress, or early frost) reduces recoverable sugar per tonne of cane processed even without a change in cane tonnage.
- **UNICA data (Sugarcane Industry Union)**: UNICA publishes biweekly crushing data for the CS region — the most watched high-frequency supply data in the No. 11 market. Published approximately 2 weeks after each reporting period, it shows: cane crushed (tonnes), sugar produced (tonnes), ethanol produced (litres, broken by hydrous and anhydrous), and the sugar/ethanol mix (how mills are allocating cane). The biweekly UNICA report is the equivalent of the EIA weekly storage report for natural gas — the primary high-frequency fundamental signal.
- **CONAB crop estimates**: Brazil's National Supply Company (CONAB) publishes monthly production estimates for sugarcane, sugar, and ethanol. The April CONAB is the first official estimate for the CS harvest that has just begun and is closely watched as the first indication of that year's crop size.

3.2 The Sugar-Ethanol Allocation Decision — The Critical Micro-Structure

Brazilian sugarcane mills can process cane into either sugar or ethanol. This flexibility is the defining feature of the Brazilian market and the most important single variable in the No. 11 micro-structure. The allocation decision is driven entirely by the relative profitability of each product:

- **The sugar-ethanol parity price**: the No. 11 price at which a Brazilian mill is indifferent between producing sugar and ethanol (in ¢/lb terms). This parity is calculated as: ethanol price at the mill gate (BRL/litre) converted to an equivalent sugar price using the cane-to-ethanol and cane-to-sugar conversion ratios and the BRL/USD exchange rate. When No. 11 trades above parity, mills prefer sugar. When No. 11 trades below parity, mills prefer ethanol.
- **Typical parity range**: varies with ethanol prices and BRL/USD. Broadly, when No. 11 is above approximately 16–18¢/lb (in many recent years), the sugar allocation is attractive. Below approximately 13–15¢/lb, ethanol is preferred. The exact parity shifts with Brazilian domestic gasoline prices (which set the effective floor for ethanol demand from flex-fuel vehicles) and the BRL/USD exchange rate.
- **Effect on the No. 11 curve**: when mills shift toward ethanol (low sugar price relative to parity), the global sugar market loses CS export supply — bullish. When mills shift toward sugar (high sugar price), the market gains CS export supply — bearish. This negative feedback mechanism makes the CS allocation a self-correcting feature of the No. 11 price at the margin.
- **Hydrous vs anhydrous ethanol**: Brazil produces two types of ethanol: hydrous (E100, used in flex-fuel cars directly) and anhydrous (blended with gasoline, E27 blend in Brazil). The split between the two depends on domestic ethanol demand and gasoline price. Mills producing more anhydrous compete more directly with gasoline and are more sensitive to Petrobras gasoline pricing policy.

3.3 Brazilian Real (BRL/USD) — The Currency Dimension

- **The BRL/USD exchange rate is a direct driver of sugar production economics**. Sugar is priced globally in USD (No. 11). Brazilian production costs are primarily in BRL (land, labour, fertiliser, energy). A weaker BRL (depreciation vs USD) means: (1) Brazilian mills earn more BRL per dollar of No. 11 revenue — increasing profitability at any given dollar price; (2) the parity price shifts: more mills find sugar attractive at a given USD sugar price because the BRL equivalent is higher; (3) the incentive to produce and export sugar increases, adding supply to the global market.
- **BRL depreciation is structurally bearish for No. 11** (more Brazilian supply at any given price). BRL appreciation is structurally bullish (Brazilian supply economics deteriorate, reducing production incentive). This relationship is not perfectly linear — it interacts with absolute price levels and parity calculation — but it is consistently documented.
- **Petrobras pricing policy**: Brazil's state oil company sets domestic gasoline prices at the refinery gate. When Petrobras raises gasoline prices, hydrous ethanol becomes more competitive in the flex-fuel market, increasing domestic ethanol demand and the parity price for sugar — potentially reducing sugar supply. When Petrobras holds gasoline prices below international parity (as occurred 2011–2015 and periodically since), domestic ethanol demand is suppressed, reducing ethanol revenue and incentivising more sugar production.

4. India — The Swing Producer and Policy Variable

India is the world's second-largest sugar producer and a market with a uniquely binary character: India swings between being a substantial net exporter and a net importer depending on its domestic harvest, and the government's decision to allow or restrict exports is one of the most market-moving events in global sugar. The timing and scale of Indian export policy decisions are the primary macro driver of No. 11 mid-curve pricing.

4.1 Indian Sugarcane Production and the Monsoon

- **Crushing season: October–March**, with Maharashtra (western India, the largest producing state), Uttar Pradesh (northern India), and Karnataka as the three primary producing states. Each state has slightly different crushing timelines and cane area response to weather.
- **Southwest Monsoon (June–September)**: the most critical weather event for Indian sugar. Adequate monsoon rainfall determines cane area and yield for the October–March crush. A weak monsoon (deficit rainfall, particularly in Maharashtra) reduces cane yield and production — the effect manifests in the following crushing season, approximately 4–8 months after a poor monsoon. El Nino years historically correlate with weaker Indian monsoons and below-average sugar production.
- **Cane area and diversion**: Indian farmers grow cane for its assured minimum price (set by the Fair and Remunerative Price, or FRP, by the central government and the State Advised Price, or SAP, by state governments). The FRP/SAP is often above market-clearing levels, creating a structural incentive to plant cane regardless of sugar market conditions. This policy intervention means Indian cane area does not respond to sugar market prices in the same way as other crops — a significant structural distortion.

4.2 Indian Export Policy — The Binary Variable

India's sugar exports are governed by central government decisions on maximum permissible export quantity (MPEQ) and subsidies. These decisions are the most market-moving policy events in the global sugar market:

- **When India's domestic production exceeds consumption** (which occurs in most years), the government faces a choice: allow unrestricted exports (bearish for No. 11), restrict exports to support domestic prices (bullish for No. 11), or subsidise exports to deplete domestic stocks (bearish for No. 11 but with WTO complications).
- **India's sugar production typically exceeds domestic consumption** by 5–10 MMT in surplus years, creating a large potential export surplus. If India exports freely, it displaces Brazilian and Thai sugar from importing markets, suppressing global prices.
- **India's domestic consumption** is approximately 25–28 MMT/year and growing steadily with population and income. This rising domestic demand gradually absorbs surplus production over time, but year-to-year, the surplus/deficit swing is the more relevant variable.
- **Government Buffer Stock Scheme**: India maintains strategic sugar stocks that can be released to the market to control domestic prices. The buffer stock level and release decisions are published by the Ministry of Food and Consumer Affairs.
- **WTO complications**: India's export subsidies for sugar have been challenged at the WTO by Brazil, Australia, and Guatemala. A 2021 WTO ruling required India to phase out production-linked subsidies. The market monitors how India complies with this ruling and whether its export support mechanisms change.

The announcement of India's export policy for a new crushing season (typically October-November) is one of the most market-moving events of the No. 11 annual calendar. A decision to allow large exports immediately adds global supply and can push No. 11 down 50–100 points (0.5–1.0¢/lb) within hours. A restriction on exports removes

supply and is immediately bullish.

5. Other Key Producers and Their Market Role

5.1 Thailand

- **Production: 8–12 MMT**, crushed November–April. Thailand is consistently the world's third-largest sugar exporter after Brazil and India. Thai cane area is sensitive to ENSO: El Nino years reduce Thailand's northeast monsoon rainfall, shrinking the crop and reducing exports. La Nina tends to benefit Thai yields.
- **Thai government export policy**: Thailand operates a two-price system — a quota A for domestic consumption and export allocation and a residual pool for discretionary export. The Thai Sugar Board manages export licences. Changes in Thai export quota allocation affect the available supply to Asian and MENA importing markets.
- **Competition with cassava**: in northeastern Thailand, farmers choose between growing cane and cassava (used for tapioca starch and ethanol). High cassava prices reduce cane area — a multi-year supply signal that affects Thai production 1–2 years forward.

5.2 European Union

- **Production: 15–18 MMT** from sugar beet, primarily in France, Germany, Poland, Belgium, Netherlands, and Austria. The EU is primarily a domestic market — it is broadly self-sufficient in sugar, occasionally exporting white sugar when production is surplus to domestic need.
- **EU sugar reform (2017)**: the abolition of EU production quotas in 2017 created larger swings in EU beet sugar production, which can now respond more freely to price signals. EU beet area and yields have become more volatile as a result.
- **Beet yield sensitivity**: EU beet yields are sensitive to summer drought (2018 and 2022 were notable drought years with materially reduced yields) and to cercospora leaf spot disease, which has spread with warmer European summers. The ban on neonicotinoid seed treatments (implemented progressively across the EU) has increased beet pest pressure and yield risk.
- **White sugar exports**: the EU exports white (refined) sugar, primarily to MENA and African markets. EU white sugar exports affect the global White Premium (London No. 5 vs ICE No. 11) and thereby influence the relative value of raw vs refined sugar trade.

5.3 China

- **Production: 10–12 MMT**, with Guangxi province (cane sugar, crushed November–April) accounting for approximately 60%, and Inner Mongolia, Xinjiang, and Yunnan (beet sugar, harvested September–November) for the remainder.
- **China is a structural net importer**, typically importing 4–6 MMT per year of raw and refined sugar under official tariff-rate quotas (TRQs) and above-quota imports. The TRQ system: China has a 1.945 MMT/year TRQ at 15% tariff; above-quota imports are tariffed at 50%. The above-quota tariff effectively sets a price at which China switches from domestic to import supply — a price ceiling that is market-relevant when No. 11 is high.
- **HFCS competition**: high-fructose corn syrup (HFCS), produced from US corn, competes with sugar in Chinese beverages and food processing. When Chinese corn prices are low (cheap HFCS), HFCS partially substitutes for sugar, reducing Chinese sugar demand. US-China trade tensions have complicated HFCS imports (25% retaliatory tariff imposed in 2018) — reducing HFCS competition and supporting Chinese sugar demand.

5.4 Australia

- **Production: 4-5 MMT**, crushed June–December in Queensland. Australia is almost entirely an export market (domestic consumption ~1 MMT). Counter-seasonal to the Northern Hemisphere, Australian raw sugar (primarily from Mackay, Bundaberg, and the Burdekin region) provides supply to Asian markets during the Northern Hemisphere off-season.

- **La Nina risk**: La Nina brings excessive Queensland rainfall which delays harvesting, reduces extraction rates, and can damage cane. El Nino brings drought and yield reduction. Both ENSO extremes carry Australian production risk.

6. Demand Fundamentals

World sugar consumption is approximately 175–185 MMT per year and grows at approximately 1.5–2.0% per year on average, driven by population growth and rising incomes in developing countries. The demand structure is relatively stable — sugar is not subject to the rapid demand dislocations that characterise energy markets — but structural changes are underway that affect the long-run price level.

6.1 Demand by Sector and Geography

- **Direct human consumption (food use)**: approximately 75–80% of global sugar use. Includes table sugar, beverages, confectionery, bakery, dairy, and processed food. This demand grows with population and GDP per capita in developing markets (income elasticity approximately +0.5 to +1.0 in low-income countries).
- **Industrial (non-food) use**: approximately 10–15% globally. Pharmaceuticals (excipients, fermentation media), chemical synthesis (bio-based chemicals), and fermentation feedstocks.
- **Biofuel (ethanol) use**: approximately 10–15% of global sugarcane output is directed to ethanol production. In Brazil this percentage is much higher (approximately 45–55% of CS cane in recent years). The ethanol sector is the primary demand competitor for Brazilian cane — it is a supply-side variable for the sugar market, not purely a demand variable.
- **Developing Asia** (India, Southeast Asia, MENA, Sub-Saharan Africa): the fastest-growing demand regions. Rising incomes drive higher sugar consumption through increased beverage, processed food, and confectionery intake. These regions account for the vast majority of incremental demand growth.
- **Developed markets** (EU, US, Australia, Japan): flat to slightly declining per capita consumption. Health consciousness, sugar taxes, and substitution toward high-intensity sweeteners (stevia, sucralose, HFCS) have plateau'd or reduced per capita sugar intake in high-income countries.

6.2 Sugar Taxes and Health Policy — The Structural Demand Headwind

- **Soft drink / sugar taxes**: the UK (implemented 2018), Mexico (2014), South Africa (2018), France, Norway, and several other jurisdictions have implemented specific taxes on sugar-sweetened beverages (SSBs). These taxes reduce SSB consumption and incentivise reformulation (lower sugar content per serving), creating a slow-moving demand headwind in affected markets.
- **Sweetener substitution**: high-fructose corn syrup (HFCS) is the primary substitute for liquid sugar in US beverages and processed food. The US No. 11 vs No. 16 price gap (driven by import protections) incentivises US manufacturers to use HFCS over cane sugar wherever possible. In global markets, stevia and other high-intensity sweeteners are gaining share in the health-conscious segment.
- **Effect on the forward curve**: these structural demand headwinds are slow-moving and affect the back end of the No. 11 curve (the long-run equilibrium price). They do not typically move the front-end spread structure which is dominated by current-year supply/demand balances.

6.3 Key Importing Countries

Importer	Import Volume	Primary Origin	Key Variable	No.11 Curve Sensitivity
China	4–6 MMT/yr	Brazil, Thailand, Australia	TRQ quota expansion; domestic production shortfall; HFCS availability	Most impactful single importer. TRQ announcements and above-quota licence decisions move mid-curve.

Indonesia	4–5 MMT/yr	Brazil, Thailand, Australia	GDP growth; domestic refinery capacity; import licensing by BULOG	Stable large importer. Import licensing (BULOG) schedule is a recurring signal.
USA	2.5–3 MMT/yr	Mexico (NAFTA/USMCA), TRQ origins	USDA TRQ allocation; domestic production shortfall; HFCS prices	Separated from No.11 market by price support; No.16 vs No.11 premium indicates US protection cost.
MENA region (Egypt, Algeria, Sudan, etc.)	5–7 MMT combined	Brazil, EU (white), India	Population growth; subsidised domestic bread/sugar programs; USD availability	Stable growing demand; white sugar preferred — affects London No.5 white premium more than No.11 directly.
Bangladesh, Pakistan, Malaysia	2–3 MMT combined	Brazil, Thailand, India	Population growth; domestic crush shortfall	Growing demand. Mostly import-dependent with limited domestic production.
Sub-Saharan Africa	3–4 MMT combined	Brazil, EU, Thailand	GDP growth; infrastructure constraints on import logistics	Growing but fragmented market. Long-term demand growth driver for global balance.

7. Trade Policy — The Most Distorted Commodity Market

Sugar is widely described as the most heavily government-intervened of all major commodity markets. Price supports, import quotas, export subsidies, and preferential trade agreements distort production in virtually every major producing and consuming country. The No. 11 contract explicitly excludes the US domestic market (which trades on No. 16 at a large premium to world prices) and prices the international free market — but this "free" market is itself influenced by policy decisions in Brazil, India, Thailand, China, the EU, and others.

7.1 US Sugar Policy

- **US farm bill price supports**: the USDA runs a price support programme for US sugar producers (both cane and beet) that effectively maintains domestic raw sugar prices well above the No. 11 world price. The "forfeiture price" (the price at which producers can forfeit sugar to the USDA in lieu of repaying government loans) acts as a floor for US domestic prices.
- **Tariff-Rate Quotas (TRQs)**: the US limits sugar imports via TRQs — a fixed quantity is allowed entry at a low tariff rate (0.625¢/lb); imports above the quota face prohibitive tariffs. The US TRQ is approximately 1.4 MMT/year for raw cane sugar plus a smaller refined sugar TRQ. TRQ allocations to origin countries (established by historical trade patterns) are occasionally adjusted.
- **USMCA (Mexico)**: under USMCA (successor to NAFTA), Mexican sugar has unlimited duty-free access to the US market in most years, subject to some sugar content rules. Mexico supplies approximately 1.5–2.0 MMT/year to the US. The USMCA sugar provisions affect the No.11 market indirectly because US import demand that is met by Mexican sugar reduces the TRQ available to No.11 origins.

7.2 Brazilian Export Subsidies and Challenges

- Brazil's government has historically provided various forms of support to the sugarcane sector (blending mandates for ethanol, financing through BNDES, regulated ethanol pricing). These supports are occasionally challenged at the WTO by competing exporters.
- Brazil does not export subsidise raw sugar directly — its cost structure is genuinely competitive globally. Brazilian cost of production is approximately 10–13¢/lb (varies with currency and input costs), making it a low-cost producer that can supply the world market at prices below most other origins' breakeven.

7.3 Preferential Trade Agreements

- **ACP (African, Caribbean and Pacific) Group — EU Everything But Arms**: developing country sugar exporters have historically had preferential access to EU markets at above-world prices. EU sugar reform (2017) and the end of EU internal support has reduced the value of these preferences as EU prices have converged closer to world levels.
- **ASEAN Free Trade Area**: within ASEAN, Thailand's sugar has preferential access to Indonesia, Malaysia, and Vietnam versus non-ASEAN origins. This affects the cost of Thai sugar for Southeast Asian importers relative to Brazilian product.

8. Weather, ENSO, and Production Risk

Weather is the most important short-cycle supply variable in sugar, operating through the monsoon-dependent production in India and Thailand and through the drought/excess rainfall risk in Brazil and Australia. ENSO is the single most powerful and predictable weather signal for the global sugar supply balance.

8.1 ENSO and Sugar Production

Country	El Nino Effect	La Nina Effect	Typical Lead Time	No.11 Impact
Brazil (CS)	Drought risk in Sao Paulo/Minas Gerais (southern producing states). Reduces cane yield and ATR. Moderate negative effect.	Excessive rainfall in southern Brazil: wet harvest, lower extraction, difficulty operating in fields. Also moderate negative, but different mechanism.	Current season; 3–6 month projection	Both ENSO extremes carry some supply risk for Brazil; neutral/moderate ENSO years tend to be best for CS production.
India	Significantly weakens the SW Monsoon (Jun–Sep). Maharashtra and Karnataka drought. Reduces cane yield for October–March crush. Historically India's worst production years align with El Nino.	Strengthens the SW Monsoon. Generally favourable for Indian cane yields (though excessive rain can damage cane in some areas).	6–12 months (monsoon determines Oct–Mar crush)	Most impactful ENSO relationship in sugar. Strong El Nino → India production deficit → export restriction → No.11 bullish. Effect on C4–C8 contracts.
Thailand	Weakens NE Monsoon (Oct–Dec). Reduced rainfall during cane growing season. Thai production typically below average in El Nino years.	Stronger than average rainfall for Thailand. Favourable yields. La Nina years tend to produce above-average Thai crops.	3–9 months	Moderate. Thai crop shortfall reduces Asian supply, supporting No.11, especially during Oct–Apr contracts.
Australia	Drought in Queensland. Reduced cane yield. Historically correlates with below-average Australian production.	Excessive rainfall: harvest delays, reduced extraction rates. Both negative — but supply disruption rather than crop failure.	Current and following season	Moderate. Counter-seasonal to NH; Australian ENSO effects felt in Jul–Dec No.11 contracts.
Cuba	Drought risk.	Usually favourable.	Current season	Smaller volume but historically relevant for the Jul contract.

8.2 The ENSO-No.11 Relationship

The most consistent ENSO-sugar price relationship is through India. Because India is large enough to swing the global balance and because its export policy is binary (on or off, not gradual), a strong El Nino that reduces Indian production and triggers an Indian export restriction creates a compound bullish shock: reduced supply and reduced export availability simultaneously. This transmission through India makes ENSO the most important long-lead fundamental signal for the No. 11 C4–C8 (mid-curve) spread positioning.

8.3 Brazil-Specific Weather Variables

- **Drought in Sao Paulo state**: the central cane-producing area of Brazil CS. A spring drought (September–November, the pre-harvest period) that follows a period of adequate rainfall actually benefits ATR (helps sucrose accumulation) but reduces cane stalk elongation (less tonnes per hectare). The balance between these two effects determines the net yield impact.
- **Frost risk in southern Sao Paulo (July–August)**: rare but acute. A severe frost event kills cane plants and can destroy portions of the standing crop. Historical frost events (1994, 1999) moved No. 11 sharply. Frost risk is geographically limited to the southernmost CS mills but market reaction can be disproportionate to the actual damage due to uncertainty and short-covering.
- **INMET and CPTEC meteorological data**: Brazil's National Institute of Meteorology (INMET) and the Center for Weather Forecasting (CPTEC) provide the primary Brazilian weather data. Commercial weather services (DTN, WeatherBell) translate these into operational sugar market signals.

9. The Forward Curve Structure — Zones and Spread Mechanics

Sugar No. 11 trades in four delivery months per year: March (H), May (K), July (N), and October (V). The gaps between contracts (October to March is 5 months; March to May is 2 months; May to July is 2 months; July to October is 3 months) create an uneven spread structure where the cost of carry between contracts varies by the number of months they span.

9.1 Carry and the No. 11 Curve Shape

- **Cost of carry for physical sugar**: warehouse storage costs (approximately $0.10–0.25/cwt/month), financing (rate-dependent), quality deterioration risk (sugar can cake or absorb moisture), and insurance. For a contango market, the spread between adjacent contracts should not exceed full carry. When it does, storage arbitrage is available (buy nearby, store, sell forward) and compresses the spread.
- **Backwardation**: when nearby contracts trade at a premium to deferred, the market is in a supply squeeze. Physical sugar is scarce relative to current demand, and holders of physical sugar can earn a premium by selling it now rather than storing. Backwardation in No. 11 is most common at the H–K (March–May) spread when Northern Hemisphere production is in seasonal trough and Brazilian production has not yet ramped.
- **The October-March spread (V–H)**: this is the largest gap between contracts (5 months). The Oct-March spread crosses the Northern Hemisphere winter off-season (minimum NH production) and the early stages of the new CS Brazilian harvest. It tends to reflect: the global supply balance entering the NH off-season, the early Brazilian harvest indicators, and the anticipated Indian export policy for the coming season.

9.2 Contract-by-Contract Fundamental Mapping

Contract	Reference Period	Primary Fundamental Driver	Secondary Driver	Key Signal
March (H)	Jan–Mar delivery. NH winter. Brazil off-season. Northern Hemisphere minimum production.	Global supply/demand balance at the beginning of the calendar year; Brazil off-season stocks and export programme; Indian export policy for current season	Thai crush progress (Nov–Mar); Australian counter-season supply to Asia	ISO quarterly balance sheet; Indian export quota announcements; Brazilian port loading data
May (K)	Mar–May delivery. Start of Brazilian CS crush. Northern Hemisphere post-winter.	Early Brazilian CS harvest indicators; CONAB April first estimate; UNICA first biweekly reports	Indian crush completion; Thai harvest conclusion; EU beet planting outlook	First UNICA biweekly report of the season; CONAB April; Brazilian port loadings
July (N)	May–Jul delivery. Brazilian harvest accelerating. Northern Hemisphere summer.	Brazilian CS mid-season crushing pace and sugar/ethanol mix; BRL/USD rate; Petrobras gasoline pricing	Australian harvest commencing; Indian monsoon early indicators; Cuban harvest completion	UNICA biweekly; BRL/USD daily; Australian BSES crop estimates; Indian monsoon onset report (IMD)
October (V)	Aug–Oct delivery. Brazilian harvest peak. Northern Hemisphere pre-winter.	Brazilian harvest volume and mix at seasonal peak; ENSO forecast for Indian monsoon and NH winter; global stocks-to-consumption ratio	Indian monsoon outcome (the Oct V contract prices the effect of the monsoon that just ended); EU beet harvest progress; USDA WASDE global balance revision	UNICA biweekly at harvest peak; USDA WASDE October revision; CPC ENSO update; Indian Ministry of Food production estimates

9.3 The White Premium — The Refinery Margin in the Curve

- **White Premium** = ICE White Sugar No. 5 price ($/MT) minus Sugar No. 11 price ($/MT equivalent). Represents the value added by refining raw sugar to white sugar.
- A wide white premium (above approximately $80–100/MT) incentivises refiners to buy raw (No. 11) and produce white. This increases demand for No. 11 from refineries in consuming regions (MENA, Asia, Europe).
- A narrow or negative white premium means refinery economics are poor. Refiners reduce intake, reducing demand for No. 11 raw sugar. Some producing countries that can export either raw or white (India, Thailand, Brazil in some cases) choose to export white when the premium is wide, reducing No. 11 deliverable supply.
- The white premium also reflects the relative balance of No. 11 raw and white refined sugar in global trade. When large importers (MENA) switch between raw and white, it directly affects the spread between the two contracts.

10. Key Data Sources

10.1 Primary Supply Data

- **UNICA (Sugarcane Industry Union, Brazil)**: biweekly crushing data for Brazil CS. Published approximately 2 weeks after each 15-day period. Data: cane crushed, sugar produced, ethanol produced, sugar/ethanol mix by state. Available at www.unica.com.br. The most important high-frequency data in the global sugar market.
- **CONAB (Brazil National Supply Company)**: monthly sugarcane, sugar, and ethanol production estimates for Brazil. The April estimate (first of the new CS season) is the most market-critical.
- **ISO (International Sugar Organization) Quarterly Report**: comprehensive global supply/demand balance, production, consumption, and trade data by country. Published quarterly. The global stocks-to-consumption ratio from ISO is the structural anchor for multi-month spread positioning.
- **USDA WASDE (World Agricultural Supply and Demand Estimates)**: monthly global sugar production, consumption, trade, and ending stocks by country. Less detailed than ISO but published more frequently (monthly) and widely referenced by US-based market participants.
- **India Ministry of Food and Consumer Affairs**: publishes monthly data on Indian sugar production (mills operating, cane crushed, sugar produced), stock estimates, and policy announcements on export quotas. Available through state sugar associations (ISMA — Indian Sugar Mills Association) and the Ministry website.
- **BSES (Bureau of Sugar Experiment Stations, Australia)** and Australian Sugar Milling Council: seasonal crop estimates and crush progress data for Australian production.
- **Thai Office of the Cane and Sugar Board (OCSB)**: Thai production and export data. Monthly production reports during the Thai crushing season (November–April).

10.2 Price and Trade Flow Data

- **Platts / S&P; Global, Argus Media**: daily physical sugar market assessments for No. 11, white sugar, and key origin differentials. The Brazilian raw sugar assessment (Sao Paulo port, ESALQ index) is the primary physical reference for CS sugar.
- **ESALQ Price Index (Escola Superior de Agricultura Luiz de Queiroz)**: the primary benchmark for Brazilian domestic raw sugar prices, published daily. The ESALQ-No. 11 basis reflects freight from Brazil to the futures delivery equivalent.
- **Brazilian port loading data (Recof-Siscomex / Secex)**: Brazilian customs (MDIC) and port logistics data showing sugar export volumes by port and destination. Weekly sugar loading data from Santos and Paranagua ports is tracked by commodity analytics services.

10.3 Currency and Macro

- **BRL/USD exchange rate**: Reuters, Bloomberg, or BCB (Banco Central do Brasil) daily. The most important currency variable in No. 11 trading. Tracked continuously as it shifts the parity calculation.
- **WTI/Brent crude oil prices**: relevant because Brazilian ethanol demand is linked to domestic gasoline prices, which are linked to crude oil. When crude oil is high, Petrobras tends to allow domestic gasoline prices to rise, increasing ethanol competitiveness and diverting cane from sugar.
- **Brazilian anhydrous ethanol price (CEPEA/ESALQ)**: the daily reference for Brazilian ethanol at the mill gate. Combined with BRL/USD, this determines the ethanol parity for the sugar-ethanol allocation decision.

11. Seasonal Patterns and the Spread Calendar

Given only four contract months, the No. 11 seasonal calendar revolves around three recurring spread opportunities that have structural fundamentals behind them.

Spread	Period Covered	Structural Driver	Typical Direction	Override Risks	Reliability
H–K (Mar–May)	Jan–Apr (3-4 months)	End of NH off-season; first weeks of Brazilian CS crush. March is the tightest supply period globally. May begins to price early Brazilian supply.	H at premium to K (backwardation) in tight years; contango in surplus years when old crop stocks are ample.	Heavy Indian export supply pressing on March. Warm winter reducing NH consumption. Early Brazilian crush start pressing May lower.	Moderate
K–N (May–Jul)	Mar–Jun (2-3 months)	Brazilian CS harvest accelerating. May prices early harvest; July prices mid-harvest. Sugar/ethanol mix decisions at mills visible by then.	K at modest premium to N (mild backwardation) when early CS season is slow or when ethanol is favoured. Contango when early harvest pace is strong.	UNICA biweekly data surprises. BRL/USD rapid move shifting parity. Petrobras gasoline price change.	Moderate
N–V (Jul–Oct)	May–Sep (3-4 months)	Brazilian CS harvest at full pace in Jul; Oct prices the outcome of the harvest + the Indian monsoon result + global stocks entering NH winter.	N at premium to V (backwardation) when harvest is below expectations or ethanol takes more cane. Contango when harvest is large and stocks building.	ENSO shift between forecast and actuals. Indian monsoon result vs forecast. Early frost scare in southern Brazil.	Moderate–High (most information-rich spread)
V–H (Oct–Mar)	Aug–Jan (5-month gap, largest spread)	Oct prices end of Brazilian harvest + opening balance for NH winter. March prices the mid-NH-winter when supply is minimum.	V at discount to H (contango) in normal supply years when Brazilian harvest was large and stocks are ample. V at premium (backwardation) when Brazilian harvest disappointed and global stocks are tight entering NH winter.	Indian export policy for coming season. ENSO forecast for NH winter. EU beet harvest result. Australian production outlook.	High — reflects the full annual supply picture; most structural of the four spreads

11.1 The Stocks-to-Consumption Ratio as the Spread Anchor

- The global sugar stocks-to-consumption (S/C) ratio — end-of-season stocks as a percentage of annual consumption — is the structural anchor for the entire No. 11 curve level and for the shape of the spread structure:
- **S/C above 45–50%**: global surplus. No. 11 tends to be in contango (ample stocks make storage economic). The spread structure is inverted to backwardation only by acute local disruptions. No. 11 flat price tends toward the cost of production of the marginal exporter.
- **S/C 35–45%**: balanced-to-tight. The curve can be either way. Weather and policy events determine the direction.
- **S/C below 30–35%**: structural tightness. No. 11 typically in backwardation across multiple spreads. Any supply disruption creates disproportionate price moves. The 2010–11 and 2022–23 periods, when global S/C fell sharply, saw No. 11 rally to 30¢/lb+ range.

12. Macro, Financial, and Cross-Market Drivers

12.1 Crude Oil and Brazilian Ethanol — The Indirect Energy Link

- As described in Section 3, the sugar-ethanol parity is mediated through Brazilian domestic gasoline prices, which are linked to international crude oil prices via Petrobras pricing policy. High crude oil = high Brazilian gasoline prices = high ethanol demand parity = mills shift to ethanol = less sugar production = bullish for No. 11. Low crude oil = low ethanol parity = mills shift to sugar = more sugar production = bearish for No. 11.
- This crude oil-sugar relationship is indirect and imperfect — it is mediated by Petrobras policy decisions, not a simple mechanical link. When Petrobras holds gasoline prices below international parity (as occurred 2011–2015), the crude-sugar link is broken because ethanol demand is artificially suppressed.

12.2 The US Dollar and Emerging Market Currencies

- **BRL/USD**: the primary currency variable, already covered extensively. A 10% BRL depreciation is roughly equivalent to a 1.5–2.5¢/lb structural shift in Brazilian supply economics.
- **INR/USD (Indian Rupee)**: affects Indian producer economics in a similar way to BRL. A weaker INR increases the rupee value of No. 11 export revenue, making Indian exports more attractive even at lower USD prices.
- **THB/USD (Thai Baht)**: similar mechanism for Thai exporters. All three emerging market currency pairs move together in risk-off scenarios (global USD strength → EM depreciation → sugar supply economics improve in all three major producers → structurally bearish for No. 11). This pattern has historically produced counter-intuitive results: global risk-off events that are bearish for most commodities can be doubly bearish for No. 11 via the currency channel.

12.3 Speculative Positioning — COT Data

- **CFTC COT for Sugar No. 11**: weekly managed money net position. Sugar is one of the most heavily speculated soft commodity futures — managed money net long positions can reach 250,000–350,000 contracts in bullish periods, creating significant crowding risk. Historically, extreme net long positions in No. 11 have preceded sharp reversals, particularly when the Brazilian harvest or Indian export news disappoints the bull thesis.
- **Commercial short (producer/processor) position**: Brazilian mills, Indian mills, and Thai mills hedge forward production by selling No. 11 futures. Heavy commercial short positioning in deferred contracts signals producer confidence in above-breakeven prices and creates known selling pressure in those months.
- **Index fund positioning**: commodity index funds (GSCI, Bloomberg Commodity Index) hold long positions in No. 11 as part of their index weightings. Index rebalancing (particularly the January roll) can create predictable mechanical buying or selling pressure in the front contracts.

12.4 Sugar Substitutes and HFCS

- High-fructose corn syrup (HFCS) is the primary substitute in beverage and processed food applications in the US and increasingly in Mexico and Asia. HFCS is made from corn (CBOT corn futures are the primary input cost). When corn prices are low, HFCS is cheap and competes with sugar in industrial food applications. When corn is expensive, HFCS loses competitiveness and sugar gains share — but this relationship is slow-moving (industrial formulations change on a contractual basis, not daily).

13. Key Data Release Schedule

Report	Publisher	Frequency	Typical Timing	Primary No.11 Curve Impact
UNICA Brazil CS Biweekly Crush Data	UNICA	Biweekly	~2 wks after reporting period (Apr–Nov)	Front 2–3 contracts during Brazilian harvest — the most important recurring data release
CONAB Monthly Production Estimate	CONAB (Brazil)	Monthly	~3rd week of month (Apr–Nov)	C2–C5 — full season production estimate; Apr report is most market-critical
ISO Quarterly Sugar Market Report	International Sugar Organization	Quarterly	~6–8 weeks after quarter end	All contracts — global S/C ratio; structural balance assessment
USDA WASDE (Sugar component)	USDA WAOB	Monthly	~11th of month, 12:00 ET	C2–C6 — global production/consumption/stocks revision; India and Brazil data most watched
ISMA Indian Production Data	Indian Sugar Mills Assoc.	Monthly	~1st week of month (Oct–Apr)	C2–C5 — Indian crush progress; most market-relevant Oct–Mar when Indian crush is active
Indian Export Quota Announcement	India Ministry of Food	As needed (typically Oct–Dec)	Variable	All contracts — one of the most market-moving single events; announced quota size directly shifts global supply outlook
NOAA ENSO forecast update	NOAA CPC / IRI	Monthly	~Mid-month	C4–C8 — seasonal temperature/rainfall probability for India, Thailand, Brazil
IMD India Monsoon Forecast	India Meteorological Dept.	Seasonal + monthly updates	Apr (first forecast), then monthly	C3–C6 — Indian production outlook; April IMD is the first monsoon signal for Oct crush
Thai OCSB Crush Data	Office of the Cane and Sugar Board	Monthly	During Thai season (Nov–Apr)	C2–C4 — Thai crop size; important for Asian regional supply
Australian BSES / ASMC Crop Estimates	BSES / Australian Sugar Milling Council	Seasonal	Pre-season (Apr) and monthly updates	C3–C5 — Australian counter-seasonal supply; relevant for Jul/Oct contracts
BRL/USD exchange rate	Reuters / Bloomberg / BCB	Daily / Continuous	Market hours	All contracts — shifts parity calculation continuously; major BRL moves are immediate market events
Brazilian ESALQ sugar/ethanol price indices	ESALQ / CEPEA	Daily	Business hours BRT	Front contracts during harvest — ethanol parity assessment for sugar/ethanol mix
Brazilian port loading data (Santos, Paranagua)	Secex / port analytics	Weekly	Variable	C1–C3 — actual export pace; confirms or contradicts UNICA production data
CFTC Commitments of Traders	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — all contracts; extreme managed money positions as contrary indicator
EU beet acreage and harvest surveys	Eurostat / Copa-Cogeca / Tallage	Seasonal	Spring (area), Autumn (harvest)	C3–C5 — EU beet crop size; affects white sugar exports and white premium

China import quota (TRQ) announcements	China MOFCOM	Annual (typically Jan–Feb) + supplemental	Variable	C2–C4 — Chinese import demand signal; above-quota licence announcements particularly market-moving

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ICE Sugar No. 11 is one of the most geopolitically complex commodity markets in the world. Supply is dominated by a handful of countries whose production decisions are shaped by weather (ENSO), government policy (India's binary export decisions, EU subsidies, US import protection), and the unique Brazilian sugar-ethanol allocation mechanism that links No. 11 directly to crude oil prices, the Brazilian real, and domestic fuel policy. Demand is growing in developing markets but facing structural headwinds in developed markets from sweetener substitution and health policy.

The forward curve's four-contract-per-year structure concentrates spread opportunities into the October-March, March-May, May-July, and July-October spreads. Of these, the October-March (V-H) spread is the most structurally informative — it crosses the annual turning point from harvest to off-season and encodes the market's assessment of the full global supply picture. The edge in No. 11 spread trading comes from accurately reading three overlapping variables simultaneously: the Brazilian CS harvest pace and sugar/ethanol mix (weekly UNICA data), the Indian production and export policy trajectory (seasonal), and the ENSO signal for both (3–9 month lead time). When all three align in the same direction, the spread move is typically large and sustained.